

Stanley Okwii

Data Scientist | Machine Learning & NLP

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PROFILE

Data scientist and ML practitioner with 6+ years' production experience and an MSc in Data Analytics (Merit, Birkbeck). I build verifiable, end-to-end systems that turn dense, unstructured documents into traceable, defensible insight — applying both traditional machine learning and large language models to surface patterns and trends for non-technical stakeholders. My recent work centres on strict output accuracy and full source traceability: every finding can be traced to the evidence behind it, the discipline an audit demands.

CORE SKILLS

Statistics & machine learning: Statistical analysis, hypothesis testing, regression, classification, SVMs; model evaluation, adversarial robustness; Pandas, NumPy, scikit-learn

NLP & generative AI: LLM integration, RAG architectures, vector search (pgvector), hybrid retrieval, deep transfer learning, hallucination mitigation

Data engineering & databases: PostgreSQL, MongoDB, Snowflake, DBT, PySpark, structured logging and monitoring, ETL pipeline design

Programming & infrastructure: Python, SQL, AWS, TypeScript; CI/CD, cloud infrastructure provisioning

EXPERIENCE

Founding Engineer (Data & NLP), Access to Energy (Stealth) – London, UK Jan 2026 – present

Built a verifiable AI research assistant that extracts factual data from dense sector reports, with strict output accuracy and full source traceability.

- Built an LLM-powered information-extraction system in which every output is traceable to its cited source document, raising measured factual accuracy by 35% over the default model response — an evidence-first design suited to audit and risk work.
- Engineered an NLP pipeline that parses, chunks and embeds hundreds of complex files concurrently, turning unstructured text into queryable vector data in minutes.
- Designed and deployed the supporting AWS infrastructure, automating environment set-up and cutting provisioning from days to under 15 minutes for secure, reproducible deployments.

Senior Full-Stack Software Engineer (Data & Backend), Harver – New York, USA Mar 2023 – May 2025

Left Harver in May 2025 to complete an MSc at Birkbeck, University of London, full-time.

- Led the backend architecture and data workflows for 3+ client-facing analytics platforms, ensuring high performance, secure data handling and maintainability.
- Partnered with product and analytics teams to scope data-heavy features, managing technical trade-offs to deliver accurate reporting to non-engineering stakeholders.
- Authored technical documentation for core data pipelines and system workflows, reducing new-hire onboarding time by 15%.

Full-Stack Software Engineer (Data & Backend), Harver – New York, USA Sept 2019 – Mar 2023

- Designed and maintained ETL pipelines that ingested and transformed candidate-assessment data, powering client scoring models and workforce-insight dashboards.
- Built analytics dashboards surfacing talent-growth and workforce-mobility trends for clients.
- Resolved database performance bottlenecks through systematic profiling, improving data-retrieval latency by 15%.

EDUCATION

MSc Birkbeck, University of London, Data Analytics – London, UK Oct 2024 – Sept 2025
Graduated with Merit. Modules included Applied Machine Learning, Big Data Analytics, Data Analytics Using R, and Neural Networks & Deep Learning.

- Dissertation on impact of adversarial training: Quantified the robustness-accuracy tradeoff of image classifiers against adversarial attacks on ResNet-50 at ImageNet scale (128k+ images, 1,000 classes).
- Built and fine-tuned CNNs using deep transfer learning (VGG, ResNet-50) in TensorFlow/Keras, and implemented PGD adversarial attacks from scratch to stress-test and retrain ResNet-50 — demonstrating practical approaches to identifying and mitigating model vulnerabilities.
- Built, tuned, and evaluated regression, classification, and SVM models using cross-validation and ROC/AUROC comparison in Python (scikit-learn) and R.

BSc Makerere University, Software Engineering – Kampala, Uganda Aug 2014 – Sept 2018
• Graduated with 2:1 (Upper Second Class Honours)